DS^{4}L: Deep Semisupervised Shared Subspace Learning for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is essential in remote sensing image analysis. The classification methods based on deep learning have attracted more and more attention. However, classification accuracy is seriously affected by the quantity of labeled data and redundant information. Therefore, a deep semisupervised shared subspace learning (DS^{4}L) model is developed to overcome these problems in this article. DS^{4}L is composed of two parts. First, the basic feature extraction (BFE) network is constructed to preliminary extract high-dimensional-space features of multiscale data and fusion them to one shared subspace. Then, a deep shared subspace learning (DSSL) network is proposed to obtain a deeper and more representative low-dimensional subspace. Moreover, to obtain a more representative subspace and alleviate dependence on labeled samples, the regular, irregular constraint, and cross-entropy (CE) loss are integrated into the model. The regular constraint is adopted to reconstruct the multiscale patches to ensure the quality of the subspace in an unsupervised manner. The irregular constraint can well embed labeled and unlabeled samples into the procedure of subspace learning (SL). Then, the CE loss is used to extract more discriminative subspace using the limited labeled samples. Finally, we perform experiments on three widely used HSI datasets. Compared with the basic SL model, the DS^{4}L's classification accuracy on the popular Salinas, Indian Pines, and PaviaU datasets are increased by 4.68%, 4.25%, and 1.57%, respectively.**

***Index Terms*— Constraint, hyperspectral image (HSI) classification, semisupervised, shared subspace learning (SL).**

I. Introduction

COMPARED with high spatial resolution and multispectral remote sensing images, the images collected from hyperspectral sensors can provide hundreds of narrow and contiguous spectral bands from the visible to the infrared spectrum. Abundant spectral information can provide a massive amount of detailed information about interesting objects that greatly improve the ability of hyperspectral images (HSIs) to recognize and distinguish ground objects. Thus, HSIs have been applied in various applications, e.g., land use, city planning, agriculture, resource exploration, and pollution detection [1]–[4].

HSI classification plays a key role in the above applications. However, the performance of HSI classification suffers from the curse of dimensionality and the lack of labeled samples [5]. To address the problem of the huge redundant information brought by highly correlated continuous bands, the related approaches have been proposed, e.g., the principal component analysis (PCA) [6], sparse graph learning [7], kernel function mapping [8], metric learning [9], band selection [10], and subspace learning (SL). Moreover, SL as an effective approach for dimensionality reduction is widely developed to reduce redundant information by transforming the high-dimensional data into the low-dimensional subspace [11]–[13]. Furthermore, the HSI classification performance is also degraded by the limited labeled data; however, the collection of labeled samples is time-consuming and labor-intensive. Therefore, designing more effective SL algorithms to extract more representative and discriminative subspace with limited samples has become an important issue in remote sensing image analysis.

With the development of artificial intelligence, deep learning-based methods have shown great potential in SL for HSIs classification. At the early stage, the stacked autoencoder [14], the variants of autoencoder [15]–[17], and the deep belief network (DBN) [18] are developed to reduce the dimensionality of HSI data to obtain the subspace for HSI classification. For example, the sparse autoencoder [19] adds a sparsity constraint on the nodes of the hidden layer to obtain a subspace representation of the input data. Manifold learning [20] is also widely used in SL, with the high-dimensional data being projected into a lower dimensional manifold representation. Especially, the convolutional neural network (CNN) [21], benefited from the local connections that can be used to learn the spatial information of the data and shared weights, are proposed to learn the discriminative feature subspace for HSIs classification [22]–[24]. However, the subspace only obtained through CNN commonly encounters poor generalization in the practical application. Thus, deep SL methods are widely studied in recent years to extract more representative and discriminative subspace [25]–[28].

Manuscript received August 22, 2021; revised September 10, 2021; accepted September 20, 2021. Date of publication October 5, 2021; date of current version January 26, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 42171388 and Grant 41801241, in part by the Fundamental Research Funds for the Central Universities under Grant 292018029 and Grant 375201906, in part by the Key Research and Development Projects of Shanxi Province under Grant 201903D121142, and in part by the Guizhou Science and Technology Plan Project under Grant Qiankehezhicheng [2020] 4Y022. *(Xudong Zhao and Li Liu contributed equally to this work.) (Corresponding author: Yuebin Wang.)*

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Digital Object Identifier 10.1109/TGRS.2021.3115354

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Graph convolution is used to reduce the dimensionality of the data as well as preserve the inherent structural information among input data, thereby enhancing the representative of the subspace [29], [30]. In [31], the deep-stacked sparse autoencoder with latent relationship embedding is used to extract the image subspace representation for HSIs classification, in which spectral–spatial graph regularization is used to extract spatial information. Besides, a multiscale adaptive sparse representation model [32] is proposed to use multiscale relevant information to complement subspace. In [33], a pixel-level and superpixel-level aware subspace learning (PSASL) method uses both pixel-level constraints and super-pixel-level constraints to obtain more representative and discriminative subspaces. Based on the above works, we continue to develop the multiscale shared subspaces. There is an issue with the range of view in the HSI recognition field. The larger size of input data brings the larger field of view, which contains different ranges of contextual information [34]. According to the theory of multiview learning, data with multiple views have a more realistic expression of real objects. Then, the concept of multiview is adopted in our model. Through multiscale input, patches of multiview are obtained. Then, a shared subspace containing more plentiful information can be obtained to improve the classification accuracy and generalization ability of the model.

Moreover, to tackle the critical problem of HSIs classification that extracting the effective subspace with the limited labeled samples, many techniques are introduced to solve this problem. Wang *et al.* [35] proposed a self-supervised HSIs classification method called hyperspectral image feature learning network (HSINet)-conditional random field (CRF) using a few labeled training samples, which combined three different levels of features, including subpixel level, pixel level, and superpixel level, to automatically extract the spectral, spatial, boundary, color, and contextual information. In [36], a semisupervised CNNs-based framework is developed for HSIs classification, which incorporates the clustering techniques into the CNNs in a self-training manner, and the spatial constraints are used to regulate the self-training process. In [37], the semisupervised method is embedded into the SL by constructing a CRF framework. Wu and Prasad [38] used abundant unlabeled samples and pseudo labels that generated by a clustering algorithm to overcome the problem of insufficient labeled samples. In [39], the subspace clustering model is used in the hypergraph-structured autoencoder for semisupervised and unsupervised classification. Among these methods, there are two ways to reduce the need for labeled samples. On the one hand, the redundant part of the data can be removed through SL, which makes more effective use of limited samples. On the other hand, by clustering the overall HSI, unlabeled samples also participate in the feedback of image features in SL. Thus, introducing unlabeled samples to the process of SL is an effective strategy for HSIs classification with limited training samples.

Among the above discussions, it can be found that neighbor pixels under a certain spatial distance usually have similar features. Moreover, these pixels can further share similar subspaces with the surrounding pixels. Then, the constraint of shared subspace can be adopted to obtain a more accurate subspace of HSI. Thus, a deep semisupervised shared subspace learning (DS^{4}L) model is developed with nearby pixels in this article. Also, considering the limited labeled samples, the regular constraint and the irregular constraint are used to ensure the quality of the obtained subspace and obtain a more representative subspace. In addition, cross-entropy (CE) loss is adopted for a more discriminative subspace.

As shown in Fig. 1, the details of the DS^{4}L can be composed of two parts, including basic feature extraction (BFE) and deep shared SL (DSSL). First, aiming to learn the shared subspace with nearby pixels of HSI, a group of convolutional layers are used to process the input multiscale patches of the initial HSI. Then, to fully utilize the spectral–spatial information of HSI, a set of convolutional layers are designed to generate the spectral–spatial features. For the second part, the initial subspace is first computed by a set of fully connected layers. The regular constraint and the irregular constraint are integrated into DSSL to ensure the quality of the obtained subspace. The multiscale patches are reconstructed by deconvolution of the feature vector to form the regular constraint. Furthermore, the irregular constraint based on the irregular superpixels can further optimize the learned subspace. Applying irregular constraint to the SL not only takes advantage of superpixels but also makes unlabeled samples participate in the training process. CE loss is calculated by the pixel-level prediction, which is performed through the softmax layer. Finally, the loss of regular constraint, the loss of irregular constraint based on superpixels, and the loss of pixel-level classification results are unified into one objective function to achieve joint training. Through experiments on several datasets, the results show that our method can surpass most semisupervised methods.

The main contributions of the article are summarized as follows.

1) An end-to-end DS^{4}L model is developed to obtain representative subspace of HSIs. The model includes two parts: the BFE network and the DSSL network.
2) The SL network integrates regular constraint and irregular constraint to enhance the representative of the subspace with the limited samples. The regular constraint fully utilizes the spectral–spatial information of multiscale HSI in the shared subspace by reconstructing multiscale patches. The irregular constraint in the SL network further optimizes the SL results with superpixel constructed by the labeled samples and unlabeled samples.
3) This semisupervised model uses labeled and unlabeled samples to train the feature learning process. The network is optimized through the above constraints and finally achieves a reliable classification result. After training and testing on several datasets, the average accuracy (AA) of DS^{4}L can reach 97.88% on the popular Salinas dataset, 95.56% on Indian Pines, and 95.73% on PaviaU.

II. Proposed Approach

In this section, the model called deep semisupervised shared subspace learning (DS^{4}L) is introduced in detail.

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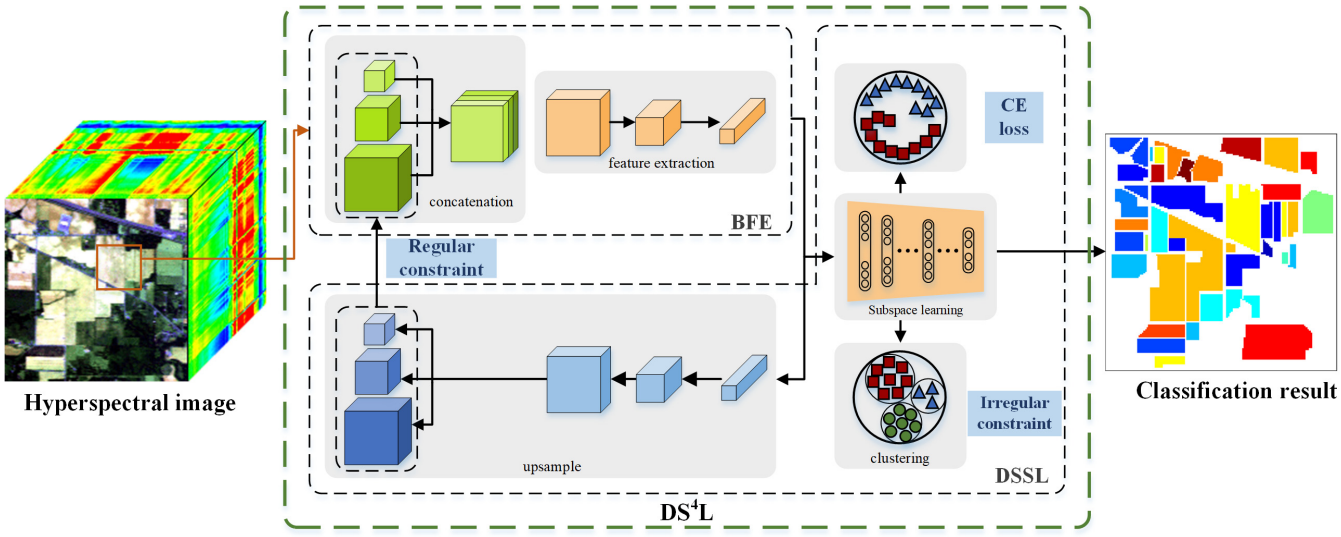


Fig. 1. Flowchart of the DS^{4}L model for HSI classification. The BFE and the DSSL are two main parts of the DS^{4}L model. Furthermore, the regular and irregular constraints and CE loss are integrated into the model to obtain representative subspace features for HSI classification.

First, the motivation of DS^{4}L is shown in Section II-A. Next, the framework of DS^{4}L is presented in Section II-B, including the BFE network and the DSSL network. Finally, the objective function and optimization of DS^{4}L are presented in Sections II-C and II-D.

A. Motivation

HSIs classification is greatly affected by the curse of dimensionality and the number of labeled samples. To overcome these problems, a DS^{4}L model is proposed. DS^{4}L is composed of a feature extraction network and SL network. First, considering utilizing similar subspace and spatial information between neighboring pixels, the multiscale feature maps containing contextual information are adopted for preliminary concatenation and extraction with the BFE network. To further eliminate redundant information with the limited training samples, the DSSL network is established. Besides, regular constraints and irregular constraints are integrated into the DSSL network to obtain a more representative subspace. To further train the network with labeled samples and evaluate the quality of the shared subspace, the subspace is fed to the softmax layer. Also, the resulting CE loss can enhance the discriminative of subspace. Finally, the objective function consisted of regular constraint, irregular constraint, and CE loss is optimized with an iterative algorithm.

B. Deep Semisupervised Shared SL

Given an HSI, this image can be transformed to an HSI dataset $\mathbf{A} \in \mathbf{R}^{w_1 \times w_2 \times d}$, where d is the number of bands and $w_1 \times w_2$ is the number of pixels. Then, the data matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n] \in \mathbf{R}^{w_1 \times w_2 \times d}$ can be built according to the HSI dataset $\mathbf{A}$. In addition, the ground truth label is represented as $\mathbf{Y} = [y_1, y_2, \ldots, y_n] \in \mathbf{R}^{w_1 \times w_2}$.

1) BFE Network: The BFE network mainly consists of multiscale inputs layer and convolutional layers. The multiscale input layer is made for processing the multiscale patches segmented from HSI to obtain a shared feature space. In detail, the pixel patches of each scale are processed to a unified scale by padding and different convolutional kernels, and then, they are concatenated to form the shared feature space. Besides that, the convolutional layers are also constructed in the BFE network to reduce dimension and extract the more abstract feature.

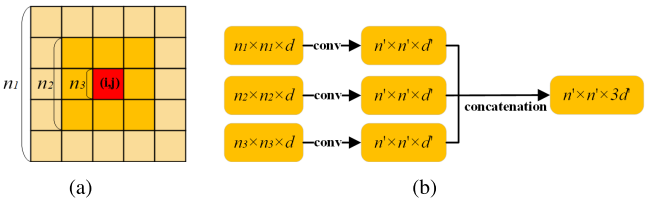


Fig. 2. Multiscale inputs process. (a) Multiscale patches. (b) Multiscale inputs are concatenated together by using different convolution kernels.

a) Multiscale Inputs: Each pixel in HSI has abundant spectral information. However, the classification accuracy is degraded by the lack of contextual information among the pixels. Thus, it is important to use pixel patches containing surrounding pixels as input data to introduce contextual information. Besides, multiscale information provides feature descriptions on different levels, which has an important effect on enriching the information in the subspace and enhancing the generalization ability of the model. Therefore, before the shared subspace is extracted, the spectral–spatial features space of the image at multiple scales should be first obtained.

The process of multiscale inputs is shown in Fig. 2. Fig. 2(a) shows the illustration of deriving the multiscale patches from HSI data. (i, j) in Fig. 2(a) is the center pixel of multiscale input patches. n_1, n_2, and n_3 denote the size of patches with different scales divided from the input HSI data. As shown in Fig. 2(b), different convolutional layers are used to process multiscale patches. The role of layers is to identify certain features from the previous multiscale patches and transform

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them to a unified scale. It is formulated as follows:

$$\left\{\begin{array}{l}\mathbf{f}_{C_1}=\psi(\mathbf{W}_{\mathbf{M_1}}\otimes\mathbf{X}_{M_1}+\mathbf{b}_{\mathbf{M_1}})\\ \mathbf{f}_{C_2}=\psi(\mathbf{W}_{\mathbf{M_2}}\otimes\mathbf{X}_{M_2}+\mathbf{b}_{\mathbf{M_2}})\\ \mathbf{f}_{C_3}=\psi(\mathbf{W}_{\mathbf{M_3}}\otimes\mathbf{X}_{M_3}+\mathbf{b}_{\mathbf{M_3}})\end{array}\right\} \tag{1}$$

where $\mathbf{X}_{M_i} \in \mathbf{R}^{n_i\times n_i\times d}$ denotes the input data, $n_i \times n_i$ represent the scale of data, d is the band of HSI. $\otimes$ represents convolution operation, and ψ is rectified linear unit (ReLU) activation function. $\mathbf{W}_{M_i}$ and $\mathbf{b}_{M_i}$ denote the weight matrix and the bias of convolutional layers in the multiscale input section, respectively. $\mathbf{f}_{C_i} \in \mathbf{R}^{n'\times n'\times d}$ denotes the output of the multiscale inputs layer. The three outputs at the same scale are concatenated to a spectral–spatial feature space $\mathbf{f}_C \in \mathbf{R}^{n'\times n'\times 3d}$.

b) Convolutional Layers: The CNN is the most representative deep structure network, which uses local connections to extract spectral–spatial feature space. The weight-share mechanism of CNN also makes the feedforward function more efficient and remarkably reduces the number of parameters. Generally, the convolutional layer can be used to extract the spectral–spatial image features. In the convolutional layers, the input patches with contextual information are convolved with a set of kernels to obtain more general and abstract spectral–spatial feature space. For the feature extraction, there are L_C convolutional layers, and the hidden layer is mainly represented as

$$\mathbf{f}_C^l=\psi\left(\mathbf{W}_{\mathbf{C}}^{\mathbf{l}}\otimes\mathbf{f}_{\mathbf{C}}+\mathbf{b}_{\mathbf{C}}^{\mathbf{l}}\right) \tag{2}$$

where $\mathbf{f}_C \in \mathbf{R}^{n'\times n'\times 3d}$ denotes the input data of the convolutional layer, which is the combination of the multiscale input, $n' \times n'$ represents the height and width of data, and $3d$ is the number of channels. $\psi(\cdot)$ is the active function and $\mathbf{f}_C^{L_C}$ is the high-dimensional-space feature vector obtained from the L_F convolutional layer. $\mathbf{W}_C^l$ and $\mathbf{b}_C^l$ denote the weight matrix and the bias of lth ($1 \le l \le L_C$) convolutional layer, respectively. The output of the previous layer is used as the next layer's input, and the final feature vector $\mathbf{f}_C^{L_C}$ is used for the next SL, superpixel clustering and reconstruction.

2) DSSL Network: As shown in Fig. 1, the DSSL network is constructed by a series of fully connected layers in which regular constraint and irregular constraint are integrated to enhance the performance of DSSL with the limited training samples.

SL can reduce the redundant information of high-dimensional features and generates low-dimensional subspace, which is more discriminative and representative. Thus, in the proposed method, the DSSL is used to transform the high-dimensional features of HSI extracted from the BFE to low-dimensional subspaces and obtain the shared subspace of deep features.

Besides, the regular constraint of DSSL preserves both spectral information and spatial correlation between HSI pixels to determine the transformation of the dimensionality reduction in the DSSL network for a more representative subspace. The feedback information in the irregular constraint is mainly obtained by unsupervised superpixel clustering, which alleviates large labeled samples necessary for HSI classification model learning.

The fully connected layers in the SL can be introduced to reduce the dimensionality and capture effective feature subspace further. The obtained low-dimensional subspace has little redundant information and strong discrimination, which can be used to enhance the classification ability of SL. The high-dimensional-space feature vector $\mathbf{f}_C^{L_C}$ is fed to the DSSL, and there are L_C fully connected layers, which can be described as

$$\mathbf{f}_S^l=\psi\left(\mathbf{W}_S^l\cdot\mathbf{f}_C^{L_C}+\mathbf{b}_S^l\right) \tag{3}$$

where $\mathbf{f}_S^l$ is the output subspace of lth ($1 \le l \le L_S$) fully connected layers, L_S is the number of fully connected layers, and $\mathbf{W}_S^l$ and $\mathbf{b}_S^l$ denote the weight matrix and the bias of the lth fully connected layers in the SL network, respectively.

a) Regular constraint: The above SL transforms the high-dimensional-space feature vector into the low-dimensional shared subspace. To evaluate the quality of the extracted shared subspace, the regular constraint is developed with deconvolution reconstructions. This constraint evaluates the information integrity with all the multiscale data to keep more important details for shared SL. Inspired by the deconvolution structure in the U-net [40], we use deconvolutional layers to reconstruct the downsampled feature maps into the high-dimensional space to evaluate the completeness of information during the SL. Compared with simple upsampling, deconvolute has more parameters that can be continuously trained, contributing to maintaining the important details.

The input $\mathbf{f}_C^{L_C}$, which obtained from the BFE network, is padded with zeros in all dimensions to generate feature maps. Then, the outputs of multiscale deconvolutional layers are obtained by using different convolutional kernels on the feature map. It is formulated as follows:

$$\left\{\begin{array}{l}\mathbf{X}_1'=\psi\left(\mathbf{W}_{D_1}^l\otimes\mathbf{f}_C^{L_C}+\mathbf{b}_{D_1}^l\right)\\ \mathbf{X}_2'=\psi\left(\mathbf{W}_{D_2}^l\otimes\mathbf{f}_C^{L_C}+\mathbf{b}_{D_2}^l\right)\\ \mathbf{X}_3'=\psi\left(\mathbf{W}_{D_3}^l\otimes\mathbf{f}_C^{L_C}+\mathbf{b}_{D_3}^l\right)\end{array}\right\} \tag{4}$$

where $\mathbf{X}_j' \in \mathbf{R}^{\mathbf{n_i}\times\mathbf{n_i}\times\mathbf{d}}$ ($j \in [1, 2, 3]$) denotes the output data, $n_i \times n_i$ represents the scale of data, L_D is the number of deconvolutional layers, and $\mathbf{W}_D^l$ and $\mathbf{b}_D^l$ denote the weight matrix and the bias of lth ($1 \le l \le L_D$) deconvolutional layers, respectively.

The feature maps with the same size as the input pixel patches in the BFE network are obtained through different convolutional kernels. Therefore, the optimization of contextual information can be evaluated by the completeness of these reconstructed feature maps. Also, the DS^{4}L model can better use the spectral–spatial information by calculating the loss between multiscale reconstructed maps and original patches. It is formulated as follows:

$$\Theta_R=\mathrm{MSE}\left(\mathbf{X}_j',\mathbf{X}_{M_j}\right)=\frac{1}{m}\sum_j^m\left(\mathbf{X}_j'-\mathbf{X}_{M_j}\right)^2 \tag{5}$$

where $\mathbf{X}_{M_j}$ is the input data of the BFE network, $\mathbf{X}_j'$ is the map reconstructed by deconvolutional layer, m is the number of maps, and Θ_R is the loss of regular constrain.

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b) Irregular constraint: To further utilize the superpixel-level information, the relationship of surrounding pixels, which often have the same category termed superpixel, is introduced to DSSL. The superpixel-level constraint information can further optimize the SL and avoid the "salt-and-pepper" problem caused by the pixel-based classification. This constraint will be automatically updated according to the optimization of the model and feedback information to the network. Besides, training a precise model for HSI classification in practical applications requires plenty of labeled training samples, which takes much time and labor. With the irregular constraint of DSSL, the training samples belonging to both labeled and unlabeled data are constructed to form the feedback information. Such feedback can train the DS^{4}L model with less labeled samples and classify the HSI in a semisupervised way.

This irregular constraint is based mainly on superpixels generated by the mean-shift clustering [41]. The specific step to fitting the mean-shift model with feature vectors $\mathbf{f}_C^{L_C}$ is creating a high-dimensional sphere. Then, offset the high-dimensional sphere according to minimize the objective function of mean shift to generate superpixels. Each superpixel obtained by the clustering model can be represented by its average features vector of similar spectral features inside. Also, the irregular constraint of the DSSL is established by the average spectral feature. The loss of the irregular constraint is represented as follows:

$$\Theta_I = \frac{1}{t}\sum_{i}^{t} \exp\left(\frac{\left\|\mathbf{f}_{S_i}^{L_S} - \overline{\mathbf{f}_{S_i}}\right\|_2^2}{\sigma}\right) \tag{6}$$

where $\mathbf{f}_{S_i}^{L_S}$ is outputted by the last layer of the deep SL belonging to the ith superpixel, $\overline{\mathbf{f}_{S_i}}$ is the average spectral feature of the ith superpixel, σ is an alternative parameter, and t is the number of the superpixel. The obtained loss of irregular constraint will be integrated into the subsequent objective function to achieve joint training of superpixel-level classification and pixel-level classification.

c) CE loss: To capture a more discriminative subspace and generate the classification result for each input sample simultaneously, the output subspace $\mathbf{f}_S^{L_S}$ of the DSSL network is connected to the softmax layer. Thus, such an end-to-end classification network can classify the HSI in a supervised way and provide an explicit objective for the network's training. The predicted label is defined as follows:

$$p\left(\hat{y} = j|\mathbf{f}_S^{L_S}\right) = \frac{\exp\left(\mathbf{f}_{S_j}^{L_S}\right)}{\sum_{i=1}^{C}\exp\left(\mathbf{f}_{S_i}^{L_S}\right)} \tag{7}$$

where $\hat{y}$ is predicted label, p is the probability of predicted label, C is the number of the category, and j denotes the category.

The CE loss that optimizes the entire network with labeled data is defined as follows:

$$\Theta_{\mathrm{CE}} = -\frac{1}{n}\sum_{i}^{n}\sum_{j=1}^{C} N(j)\log\left(p_i\left(\hat{y}_i = j|\mathbf{f}_S^{L_S}\right)\right) \tag{8}$$

where n is the number of predicted pixels. $N(j)$ is 1 when j is consistent with the predicted label $\hat{y}_j$ of pixel i ($1 \le i \le n$); otherwise, the value is 0.

C. Overall Objective Function

The objective function of DS^{4}L can be formed with the three constraints obtained from (5), (6), and (8)

$$\begin{aligned}\Theta &= \lambda_1\Theta_R + \lambda_2\Theta_I + \lambda_3\Theta_{\mathrm{CE}} \\ &= \lambda_1\left[\frac{1}{m}\sum_{j}^{m}\left(\mathbf{X}'_j - \mathbf{X}_{M_j}\right)^2\right] \\ &\quad + \lambda_2\left[\frac{1}{t}\sum_{i}^{t}\exp\left(\frac{\left\|\mathbf{f}_{S_i}^{L_S} - \overline{\mathbf{f}_{S_i}}\right\|_2^2}{\sigma}\right)\right] \\ &\quad + \lambda_3\left[-\frac{1}{n}\sum_{i}^{n}\sum_{j=1}^{C} N(j)\log\left(p_i\left(\hat{y}_i = j|\mathbf{f}_S^{L_S}\right)\right)\right]\end{aligned} \tag{9}$$

where λ_1, λ_2, and λ_3 are the parameters to balance the weight of the regular constraint, irregular constraint, and CE loss, respectively, M_j is the scale of the input data, and n is the number of predicted pixels.

D. Optimization of DS^{4}L

The loss function is constructed through three constraints. The Adam optimizer and the backpropagation learning are adopted in this framework. Through the backpropagation, the weights and biases of each layer are updated. The other adjustable parameters are changed based on each experiment's results until the model has the best classification accuracy. To conveniently calculate the derivative of the Θ_I to $\mathbf{W}$ and $\mathbf{b}$, the lth fully connected layer in the SL can be expressed as

$$\begin{cases}\mathbf{z}_S^{l+1} = \mathbf{W}_S^{l+1}\mathbf{f}_S^l + \mathbf{b}_S^{l+1} \\ \mathbf{f}_S^{l+1} = \psi\left(\mathbf{z}_S^{l+1}\right)\end{cases} \tag{10}$$

where $\mathbf{z}_S^{l+1}$ is the output without the active function.

An intermediate parameter δ^l that denotes the derivative of Θ_I to $\mathbf{z}_S$ of each lth layer can be defined as

$$\delta_S^{l+1} = \frac{\partial\Theta_I}{\partial\mathbf{z}_S^{l+1}} = \frac{\partial\Theta_I}{\partial\mathbf{f}_S^{l+1}}\cdot\frac{\partial\mathbf{f}_S^{l+1}}{\partial\mathbf{z}_S^{l+1}} = \frac{2}{\sigma}\left(\mathbf{f}_S^{l+1} - \bar{\mathbf{f}}\right)\odot\psi'\left(\mathbf{z}_S^{l+1}\right) \tag{11}$$

where $\odot$ denotes the element-wise operation.

Thus, the derivative of Θ_S to $\mathbf{W}$ and $\mathbf{b}$ in the lth layer can be calculated as

$$\frac{\partial\Theta_I}{\partial\mathbf{W}_S^{l+1}} = \frac{\partial\Theta_I}{\partial\mathbf{z}_S^{l+1}}\cdot\frac{\partial\mathbf{z}_S^{l+1}}{\partial\mathbf{W}_S^{l+1}} = \delta_S^{l+1}\left(\mathbf{f}_S^l\right)^T \tag{12}$$

$$\frac{\partial\Theta_I}{\partial\mathbf{b}_S^{l+1}} = \frac{\partial\Theta_I}{\partial\mathbf{z}_S^{l+1}}\cdot\frac{\partial\mathbf{z}_S^{l+1}}{\partial\mathbf{b}_S^{l+1}} = \delta_S^{l+1}. \tag{13}$$

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Also, the parameters' derivative of the next layer can be obtained

$$\delta_S^l = \frac{\partial \Theta_I}{\partial \mathbf{z}_S^l} = \frac{\partial \Theta_I}{\partial \mathbf{z}_S^{l+1}} \cdot \frac{\partial \mathbf{z}_S^{l+1}}{\partial \mathbf{z}_S^l} = \left(\mathbf{W}_S^{l+1}\right)^T \delta_S^{l+1} \odot \psi'\left(\mathbf{z}_S^l\right) \quad (14)$$

$$\frac{\partial \Theta_I}{\partial \mathbf{W}_S^l} = \frac{\partial \Theta_I}{\partial \mathbf{z}_S^l} \cdot \frac{\partial \mathbf{z}_S^l}{\partial \mathbf{W}_S^l} = \delta_S^l \left(\mathbf{f}_S^{l-1}\right)^T \quad (15)$$

$$\frac{\partial \Theta_I}{\partial \mathbf{b}_S^l} = \frac{\partial \Theta_I}{\partial \mathbf{z}_S^l} \cdot \frac{\partial \mathbf{z}_S^l}{\partial \mathbf{b}_S^l} = \delta_S^l. \quad (16)$$

Similarly, we can obtain the parameters' derivative of the BFE as follows:

$$\delta_C^l = \left(\mathbf{W}_S^l\right)^T \delta_S^l \odot \psi'\left(\mathbf{z}_C^l\right) \quad (17)$$

$$\frac{\partial \Theta_I}{\partial \mathbf{W}_C^l} = \delta_C^l \left(\mathbf{f}_C^{l-1}\right)^T \quad (18)$$

$$\frac{\partial \Theta_I}{\partial \mathbf{b}_C^l} = \delta_C^l. \quad (19)$$

To conveniently obtain the derivative of the Θ_R to convolution kernel $\mathbf{W}$ and bias $\mathbf{b}$, the L_D deconvolution layer of each scale can be expressed in the matrix form

$$\begin{cases} \mathbf{z}_D^{l+1} = \mathbf{W}_D^{l+1} \otimes \mathbf{f}_D^l + \mathbf{b}_D^{l+1} \\ \mathbf{X}'_D = \psi\left(\mathbf{z}_D^{l+1}\right) \end{cases} \quad (20)$$

where the final outputs are denoted by $\mathbf{X}_{D_j}^l$, which are the matrix forms corresponding to $\mathbf{X}'_j$ ($j \in [1, 2, 3]$) in formula (6). $\mathbf{W}_D^{l+1}$ and $\mathbf{b}_D^{l+1}$ are the convolution kernel and bias at the L_Dth layer (and $1 \le l < L_D$), respectively. $\mathbf{z}_D^l$ is the output matrix without the active function.

The derivative of Θ_R to $\mathbf{z}_D$ of the lth deconvolution layer can be defined as

$$\delta_D^{l+1} = \frac{\partial \Theta_R}{\partial \mathbf{z}_D^{l+1}} = \frac{\partial \Theta_R}{\partial \mathbf{X}'_D} \cdot \frac{\partial \mathbf{X}'_D}{\partial \mathbf{z}_D^{l+1}} = 2\left(\mathbf{X}' - \mathbf{X}_M\right) \odot \psi'\left(\mathbf{z}_D^{l+1}\right). \quad (21)$$

Thus, the derivative of Θ_R to $\mathbf{W}$ and $\mathbf{b}$ in the lth layer can be calculated as

$$\frac{\partial \Theta_R}{\partial \mathbf{W}_D^{l+1}} = \frac{\partial \Theta_R}{\partial \mathbf{z}_D^{l+1}} \cdot \frac{\partial \mathbf{z}_D^{l+1}}{\partial W_D^{l+1}} = \delta_D^{l+1} \otimes \mathbf{f}_D^l \quad (22)$$

$$\frac{\partial \Theta_R}{\partial \mathbf{b}_D^{l+1}} = \frac{\partial \Theta_R}{\partial \mathbf{z}_D^{l+1}} \cdot \frac{\partial \mathbf{z}_D^{l+1}}{\partial \mathbf{b}_D^{l+1}} = \sum_x \sum_y \delta_D^{l+1}. \quad (23)$$

The derivative of Θ_R to $\mathbf{W}$ and $\mathbf{b}$ in the next layer can be obtained

$$\delta_D^l = \frac{\partial \Theta_R}{\partial \mathbf{z}_D^l} = \delta_D^{l+1} \otimes \mathrm{ROT}\,180\left(\mathbf{W}_D^{l+1}\right) \odot \psi'\left(\mathbf{z}_D^l\right) \quad (24)$$

$$\frac{\partial \Theta_R}{\partial \mathbf{W}_D^l} = \frac{\partial \Theta_R}{\partial \mathbf{X}_D^l} \cdot \frac{\partial \mathbf{X}_D^l}{\partial \mathbf{W}_D^l} = \delta_D^l \otimes \mathbf{f}_D^{l-1} \quad (25)$$

$$\frac{\partial \Theta_R}{\partial \mathbf{b}_D^l} = \frac{\partial \Theta_R}{\partial \mathbf{z}_D^l} \cdot \frac{\partial \mathbf{z}_D^l}{\partial \mathbf{b}_D^l} = \sum_x \sum_y \delta_D^l. \quad (26)$$

Similarly, we can obtain the parameters' derivative of the BFE as follows:

$$\delta_{C_2}^l = \delta_D^l \otimes \mathrm{ROT}\,180\left(\mathbf{W}_D^l\right) \odot \psi'\left(\mathbf{z}_C^l\right) \quad (27)$$

$$\frac{\partial \Theta_R}{\partial \mathbf{W}_C^l} = \delta_{C_2}^l \left(\mathbf{f}_C^{l-1}\right)^T \quad (28)$$

$$\frac{\partial \Theta_R}{\partial \mathbf{b}_C^l} = \delta_{C_2}^l. \quad (29)$$

Both the BFE network and the DSSL network are optimized by Θ_{CE}. Thus, there is the derivation process of DSSL parameters, and the gradient of other parameters in BFE can be obtained in the same way.

To obtain the derivative of Θ_{CE} to convolution kernel $\mathbf{W}_S^{l+1}$ and $\mathbf{b}_S^{l+1}$ at the L_Sth layer (and $1 \le l < L_S$), the L_Sth layer of the network can be expressed in the matrix form

$$\begin{cases} \mathbf{z}_S^{l+1} = \mathbf{W}_S^{l+1} \mathbf{f}_S^l + \mathbf{b}_S^l \\ \mathbf{f}_S^{l+1} = \varphi\left(\mathbf{z}_S^{l+1}\right) \end{cases} \quad (30)$$

where φ denotes the softmax activation function.

The derivative of Θ_{CE} to $\mathbf{z}_S^{l+1}$ can be defined as

$$\delta_{\mathrm{CE}}^{l+1} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^{l+1}} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{f}_S^{l+1}} \cdot \frac{\partial \mathbf{f}_S^{l+1}}{\partial \mathbf{z}_S^{l+1}} = \left(\mathbf{f}_S^{l+1} - Y\right). \quad (31)$$

Thus, the derivative of Θ_{CE} to $\mathbf{W}$ and $\mathbf{b}$ in the softmax layer can be calculated as

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{W}_S^{l+1}} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^{l+1}} \cdot \frac{\partial \mathbf{z}_S^{l+1}}{\partial \mathbf{W}_S^{l+1}} = \delta_{\mathrm{CE}}^{l+1} \left(\mathbf{f}_S^l\right)^T \quad (32)$$

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{b}_S^{l+1}} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^{l+1}} \cdot \frac{\partial \mathbf{z}_S^{l+1}}{\partial \mathbf{b}_S^{l+1}} = \delta_{\mathrm{CE}}^{l+1}. \quad (33)$$

The next fully connected layer in the DSSL network can be expressed as

$$\begin{cases} \mathbf{z}_S^l = \mathbf{W}_S^l \mathbf{f}_S^{l-1} + \mathbf{b}_S^l \\ \mathbf{f}_S^l = \psi\left(\mathbf{z}_S^l\right) \end{cases} \quad (34)$$

where ψ denotes the activation function (and the $1 \le l < L_S$).

Thus, the derivative of Θ_{CE} to $\mathbf{W}$ and $\mathbf{b}$ in the next fully connected layer can be calculated as

$$\delta_{\mathrm{CE}}^l = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^l} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^{L_S}} \cdot \frac{\partial \mathbf{z}_S^{L_S}}{\partial \mathbf{z}_S^l} = \left(\mathbf{W}_S^{l+1}\right)^T \delta_{\mathrm{CE}}^{l+1} \odot \psi'\left(\mathbf{z}_S^l\right) \quad (35)$$

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{W}_S^l} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^l} \cdot \frac{\partial \mathbf{z}_S^l}{\partial \mathbf{W}_S^l} = \delta_{\mathrm{CE}}^l \left(\mathbf{f}_S^{l-1}\right)^T \quad (36)$$

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{b}_S^l} = \frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{z}_S^l} \cdot \frac{\partial \mathbf{z}_S^l}{\partial \mathbf{b}_S^l} = \delta_{\mathrm{CE}}^l. \quad (37)$$

Similarly, we can obtain the parameters' derivative of the BFE as follows:

$$\delta_{C_3}^l = \left(\mathbf{W}_C^l\right)^T \delta_{\mathrm{CE}}^l \odot \psi'\left(\mathbf{z}_C^l\right) \quad (38)$$

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{W}_C^l} = \delta_{C_3}^l \left(\mathbf{f}_C^{l-1}\right)^T \quad (39)$$

$$\frac{\partial \Theta_{\mathrm{CE}}}{\partial \mathbf{b}_C^l} = \delta_{C_3}^l. \quad (40)$$

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1) Optimization for BFE Network: Adam is used to calculate the influence on $\mathbf{W}$ and $\mathbf{b}$ of each part separately. $\mathbf{W}$ is optimized by minimizing the loss value when $\mathbf{b}$ is fixed. The weight matrix $\mathbf{W}_C^l$ and the bias $\mathbf{b}_C^l$ of L_C convolutional layer in BFE are optimized by Θ_R, Θ_I, and Θ_{CE}

$$\mathbf{W}_C^l = \mathbf{W}_C^l - r\left(\frac{\partial\Theta_R}{\partial\mathbf{W}_C^l} + \frac{\partial\Theta_I}{\partial\mathbf{W}_C^l} + \frac{\partial\Theta_{\mathrm{CE}}}{\partial\mathbf{W}_C^l}\right)$$
$$\mathbf{b}_C^l = \mathbf{b}_C^l - r\left(\frac{\partial\Theta_R}{\partial\mathbf{b}_C^l} + \frac{\partial\Theta_I}{\partial\mathbf{b}_C^l} + \frac{\partial\Theta_{\mathrm{CE}}}{\partial\mathbf{b}_C^l}\right) \tag{41}$$

where r is the learning rate.

2) Optimization for DSSL Network: The weight matrix $\mathbf{W}_S^l$ and the bias $\mathbf{b}_S^l$ of the DSSL network are optimized by Θ_S and Θ_{CE}

$$\mathbf{W}_S^l = \mathbf{W}_S^l - r\left(\frac{\partial\Theta_I}{\partial\mathbf{W}_S^l} + \frac{\partial\Theta_{\mathrm{CE}}}{\partial\mathbf{W}_S^l}\right)$$
$$\mathbf{b}_S^l = \mathbf{b}_S^l - r\left(\frac{\partial\Theta_I}{\partial\mathbf{b}_S^l} + \frac{\partial\Theta_{\mathrm{CE}}}{\partial\mathbf{b}_S^l}\right). \tag{42}$$

The weight matrix $\mathbf{W}_D^l$ and the bias $\mathbf{b}_D^l$ of the deconvolution layers are optimized by Θ_R

$$\mathbf{W}_D^l = \mathbf{W}_D^l - r\frac{\partial\Theta_R}{\partial\mathbf{W}_D^l}$$
$$\mathbf{b}_D^l = \mathbf{b}_D^l - r\frac{\partial\Theta_R}{\partial\mathbf{b}_D^l}. \tag{43}$$

The superpixel parameters that are irregularly constrained will also be optimized with the change of the feature vector output by the BFE.

III. Experimental Results

In this section, the performance of the proposed model is evaluated on three standard datasets. Subsequently, the result proves that DS^{4}L can effectively improve the classification accuracy by comparing DS^{4}L with related semisupervised methods.

A. Datasets

Test the effectiveness of the model in the following three datasets.

1) Salinas Scene: The airborne visible infrared imaging spectroradiometer (AVIRIS) sensor collects this scene over Salinas Valley. The 224 bands in this image contain 20 water absorption bands, which can be discarded. The image's size is 512 × 217, of which 54 129 pixels can be applied to classification. These pixels contain 16 types of vegetables, bare soil, and fallow.

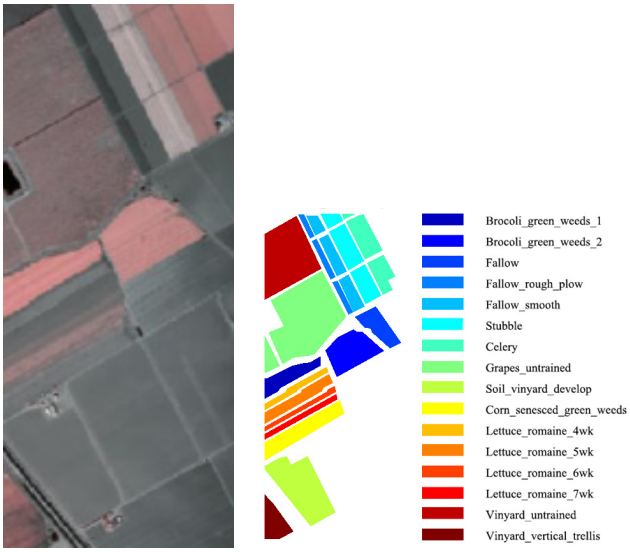
2) Indian Pines: This scene is gathered by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana and consisted of 145 × 145 pixels and 224 spectral. The Indian Pines are divided into 16 types of objects.

3) Pavia University Scene: Pavia University is the hyperspectral data of the image made by the reflective optics system imaging spectrometer (ROSIS) sensor on Pavia city in Italy. The number of useful spectral bands is 103 for this Pavia University. Also, the size of the image is 610 × 340, which contains the background and nine types of features.

TABLE I

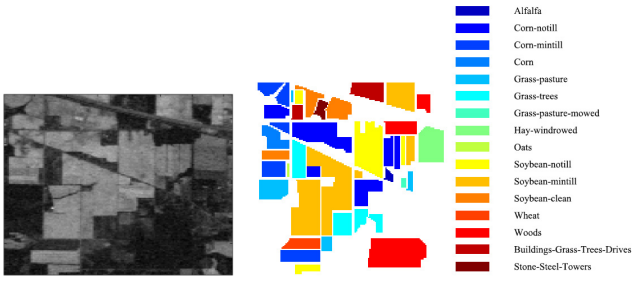
Round Truth Classes for the Salinas Scene and Their Respective Sample Number

	Class	Training Samples	Testing Samples
1	Brocoli_green_weeds_1	100	1909
2	Brocoli_green_weeds_2	100	3626
3	Fallow	100	1876
4	Fallow_rough_plow	100	1294
5	Fallow_smooth	100	2578
6	Stubble	100	3859
7	Celery	100	3479
8	Grapes_untrained	100	11171
9	Soil_vinyard_develop	100	6103
10	Corn_senesced_green_weeds	100	3178
11	Lettuce_romaine_4wk	100	968
12	Lettuce_romaine_5wk	100	1827
13	Lettuce_romaine_6wk	100	816
14	Lettuce_romaine_7wk	100	970
15	Vinyard_untrained	100	7168
16	Vinyard_vertical_trellis	100	1707
	Total	1600	52529



(a) (b)

Fig. 3. Salinas dataset. (a) False RGB image. (b) Ground-truth map.



(a) (b)

Fig. 4. Indian Pines dataset. (a) False RGB image. (b) Ground-truth map.

The detailed information of training samples and test samples used in experiment, which can be seen in Tables I–III. The false RGB images and ground-truth maps of these three dataset are shown in Figs. 2–4.

B. Experiment Implementation

In the BFE network, the number of 1, 3, and 5 is used as the input scales n_i in this article, and the multiscale input layer

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TABLE II

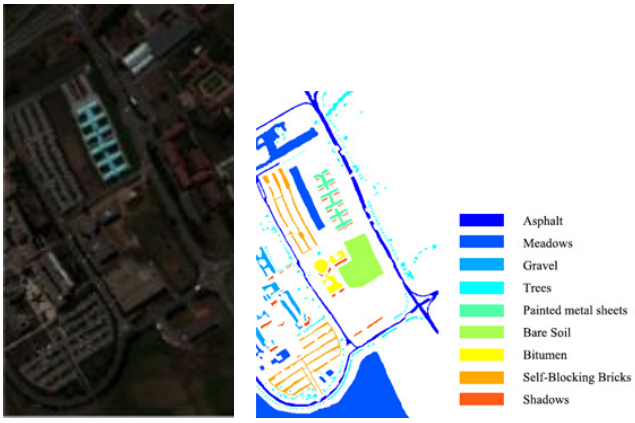
GROUND TRUTH CLASSES FOR THE INDIAN PINES SCENE AND THEIR RESPECTIVE SAMPLE NUMBER

	Class	Training Samples	Testing Samples
1	Alfalfa	23	23
2	Corn-notill	50	1378
3	Corn-mintill	50	780
4	Corn	50	187
5	Grass-pasture	50	433
6	Grass-trees	50	680
7	Grass-pasture-mowed	14	14
8	Hay-windrowed	50	428
9	Oats	10	10
10	Soybean-notill	50	922
11	Soybean-mintill	50	2405
12	Soybean-clean	50	543
13	Wheat	50	155
14	Woods	50	1215
15	Buildings-Grass-Trees-Drives	50	336
16	Stone-Steel-Towers	46	47
	Total	693	9556

TABLE III

GROUND TRUTH CLASSES FOR THE PAVIA UNIVERSITY SCENE AND THEIR RESPECTIVE SAMPLE NUMBER

	Class	Training Samples	Testing Samples
1	Asphalt	100	6531
2	Meadows	100	18549
3	Gravel	100	1999
4	Trees	100	2964
5	Painted metal sheets	100	1245
6	Bare Soil	100	4929
7	Bitumen	100	1230
8	Self-Blocking Bricks	100	3582
9	Shadows	100	847
	Total	900	41876



(a) (b)

Fig. 5. PaviaU dataset. (a) False RGB image. (b) Ground-truth map.

got $1 \times 1 \times d$, $3 \times 3 \times d$, and $5 \times 5 \times d$ patches for each central pixel. Different padding and convolution kernels are used to process these patches. Then, three $5 \times 5 \times 256$ patches are concatenated to one spectral–spatial feature map with the scale of $5 \times 5 \times 768$ for other convolutional layers. Finally, the high-dimensional-space feature vector $1 \times d$ is obtained after the BFE network.

Each networks' number of layers L_C, L_S, and L_D are equal to 3, 5, and 3, respectively. The weights and biases of each layer in the network are initialized randomly. Other parameters are adjusted in a suitable range. For example, in the DSSL network, the mapping dimension of output subspace is selected from the set {10, 20, 30, 40, 50, 60, 70, 80}.

C. Comparisons With Related Approaches

The following approaches are adopted to compare with our proposed DS^{4}L in the field of HSI classification accuracy.

1) PCA + Label Propagation [42]: The label propagation (LP) classification model works by constructing a similarity graph on the input data and propagating the known labels to the unlabeled pixels.

2) Low-Rank Representation [43] + Softmax: The matrix's low-rank representation (LRR) represents the dataset matrix as a linear combination of the base matrix and makes the coefficient matrix of the linear combination lowest rank representation.

3) Kernel-PCA (KPCA) + CE [44]: Kernel-PCA (KPCA), as a nonlinear PCA, uses the kernel function to calculate the covariance matrix and maps the original spatial data to the high-dimensional space for dimensionality reduction, thereby extracting nonlinear subspace.

4) Semisupervised 2-D CNN (2DCNN) [45]: Semisupervised CNN can automatically extract the spectral–spatial information of images through convolutional layers. In addition, a skip connection parameter is added between the encoder layer and the decoder layer to solve the problem of limited labeled samples in a semisupervised way.

5) Multiscale 3-D CNN (3DCNN) [46]: The 3-D convolutional network in this method is used to obtain the 2-D spatial information and 1-D spectral information synchronously for joint learning.

6) Graph Convolutional Network [47]: The nonlinear and discriminative spectral–spatial features extracted by graph convolutional network (GCN) are used to replace traditional linear features, which makes more accurate classification with limited labeled samples.

The above widely use methods are used to compare our DS^{4}L. PCA + LP uses radial basis function (RBF) as the kernel function, the Gaussian distribution's width, and the model's complexity are adjusted by gamma, which is selected from $\{1, 2, 3, \ldots, 13, 14, 15\}$. KPCA + CE calculates the distance between each sample through $\exp(-\gamma \, ||x_i - x_j||^2)$, where γ represents the heat kernel. The number of dimensions in KPCA + CE and LRR is chosen from the set {10, 20, 30, 40, 50, 60, 70, 80}. To prove that our method improves the classification effect, we also used the semisupervised CNN, Multiscale 3DCNN, and GCN methods to compare DS^{4}L.

D. Experimental Results

The AA, overall accuracy (OA), and kappa coefficient (Kappa) are used in our experiment to evaluate the effectiveness of the proposed classification model. Tables IV–VI summarize the classification results of each algorithm and the corresponding evaluation indicators.

Based on the results listed in Tables IV–VII and the observations in Figs. 6–8, there are the following conclusions.

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TABLE IV
HSI CLASSIFICATION RESULTS BY SELECTING 2% LABELED SAMPLES FOR EACH CLASS ON THE SALINAS DATASET

Land Cover Type	PCA+LP	LRR+softmax	KPCA+CE	2DCNN	3DCNN	GCN	DS^{4}L
Brocoli_green_weeds_1	98.01	98.17	99.21	99.90	60.81	98.59	99.42
Brocoli_green_weeds_2	98.92	98.43	97.77	99.97	99.23	99.67	99.81
Fallow	95.90	95.63	97.97	98.35	99.63	98.24	100.00
Fallow_rough_plow	99.46	96.45	99.69	100.00	61.80	99.23	99.77
Fallow_smooth	96.63	94.88	97.52	98.68	95.73	98.80	98.10
Stubble	99.72	98.70	95.75	99.64	99.95	99.77	100.00
Celery	99.54	99.60	99.63	99.86	76.96	99.66	99.74
Grapes_untrained	66.17	65.20	78.18	79.17	93.52	82.22	87.03
Soil_vinyard_develop	98.77	96.13	87.24	99.33	99.85	99.69	99.89
Corn_senesced_green_weeds	88.48	92.13	93.68	94.84	95.81	92.39	97.26
Lettuce_romaine_4wk	92.36	95.35	96.28	98.97	99.80	97.62	99.48
Lettuce_romaine_5wk	99.78	92.12	98.47	100.00	99.78	99.89	100.00
Lettuce_romaine_6wk	99.02	95.47	99.51	100.00	99.64	99.02	99.63
Lettuce_romaine_7wk	94.85	87.22	99.07	97.63	94.04	92.47	99.79
Vinyard_untrained	64.22	60.60	76.83	73.51	80.70	62.30	87.47
Vinyard_vertical_trellis	92.91	97.25	94.08	99.24	99.59	98.07	98.65
AA	92.80	91.46	94.43	96.19	91.05	94.85	**97.88**
OA	86.07	84.75	89.27	91.31	91.41	90.06	**95.14**
KAPPA	0.84	0.83	0.88	0.90	0.90	0.89	**0.95**

TABLE V
HSI CLASSIFICATION RESULTS BY SELECTING 2% LABELED SAMPLES FOR EACH CLASS ON THE INDIAN PINES DATASET

Land Cover Type	PCA+LP	LRR+softmax	KPCA+CE	2DCNN	3DCNN	GCN	DS^{4}L
Alfalfa	64.29	92.86	92.86	100.00	100.00	87.50	100.00
Corn-notill	47.06	59.19	96.88	82.98	76.20	68.00	90.29
Corn-mintill	60.96	50.96	91.28	81.92	83.72	76.85	88.36
Corn	76.64	65.69	97.86	95.62	90.91	83.94	100.00
Grass-pasture	85.90	88.77	96.54	96.61	97.23	86.42	97.39
Grass-trees	90.95	92.54	90.59	98.10	92.35	90.00	99.52
Grass-pasture-mowed	75.00	100.00	100.00	100.00	85.71	100.00	92.86
Hay-windrowed	90.74	94.44	99.77	99.74	100.00	91.27	100.00
Oats	66.67	66.67	100.00	100.00	100.00	71.43	100.00
Soybean-notill	69.04	51.03	72.02	86.01	95.55	79.36	84.75
Soybean-mintill	57.88	59.53	75.88	78.68	90.06	55.50	93.29
Soybean-clean	55.38	66.53	98.34	82.76	75.69	66.33	94.52
Wheat	97.14	98.10	98.71	100.00	98.06	99.05	100.00
Woods	78.80	82.58	90.12	92.36	98.60	90.82	98.20
Buildings-Grass-Trees-Drives	46.50	65.38	54.46	89.16	93.15	71.68	89.86
Stone-Steel-Towers	100.00	88.89	100.00	100.00	100.00	96.77	100.00
AA	72.68	76.45	90.96	92.75	92.33	82.18	**95.56**
OA	65.68	67.33	86.17	86.34	89.61	73.65	**93.35**
KAPPA	0.61	0.63	0.84	0.84	0.88	0.70	**0.92**

TABLE VI
HSI CLASSIFICATION RESULTS BY SELECTING 2% LABELED SAMPLES FOR EACH CLASS ON THE PAVIAU DATASET

Land Cover Type	PCA+LP	LRR+Softmax	KPCA+CE	2DCNN	3DCNN	GCN	DS^{4}L
Asphalt	69.04	62.32	83.76	89.25	79.96	82.01	92.22
Meadows	76.75	80.44	87.48	98.24	98.07	90.98	96.45
Gravel	79.64	71.37	95.89	85.29	97.70	73.29	92.35
Trees	89.54	85.79	59.75	97.91	94.06	92.61	97.81
Painted metal sheets	99.20	98.66	95.74	99.92	99.60	99.28	100.00
Bare Soil	74.01	64.58	85.96	90.99	99.55	83.53	92.17
Bitumen	92.85	76.02	93.25	89.59	99.11	92.52	96.34
Self-Blocking Bricks	77.36	63.11	89.53	89.39	84.31	88.30	94.19
Shadows	100.00	96.49	68.59	100.00	96.93	100.00	100.00
AA	84.27	77.65	84.44	93.40	94.37	89.17	**95.73**
OA	77.93	74.92	85.37	94.42	93.99	88.22	**95.17**
KAPPA	0.72	0.67	0.81	0.93	0.92	0.84	**0.94**

1) By observing the classified images, we find that the classification of Vinyard_untrained in Salinas and the Meadows in PaviaU is obviously more accurate. The DS^{4}L method effectively reduces the misclassification of adjacent pixels. It shows that the SL with regular and irregular constraints effectively alleviates the "salt-and-pepper" problem and pixel-level misclassification. In conclusion, the image results show that the DS^{4}L method significantly improves the accuracy of the HSI classification.

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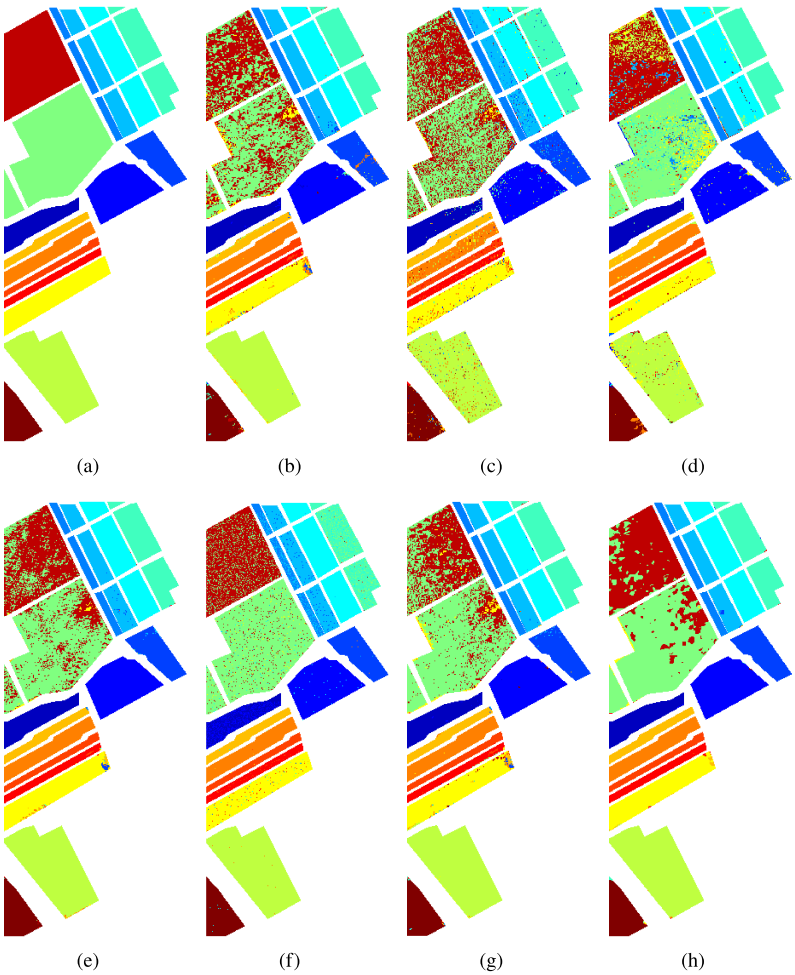


Fig. 6. Classification maps of the Salinas dataset by selecting 2% labeled samples for each class. Classification map obtained by (a) ground truth, (b) PCA + LP, (c) LRR + softmax, (d) KPCA + CE, (e) 2DCNN, (f) 3DCNN, (g) GCN, and (h) DS^{4}L.

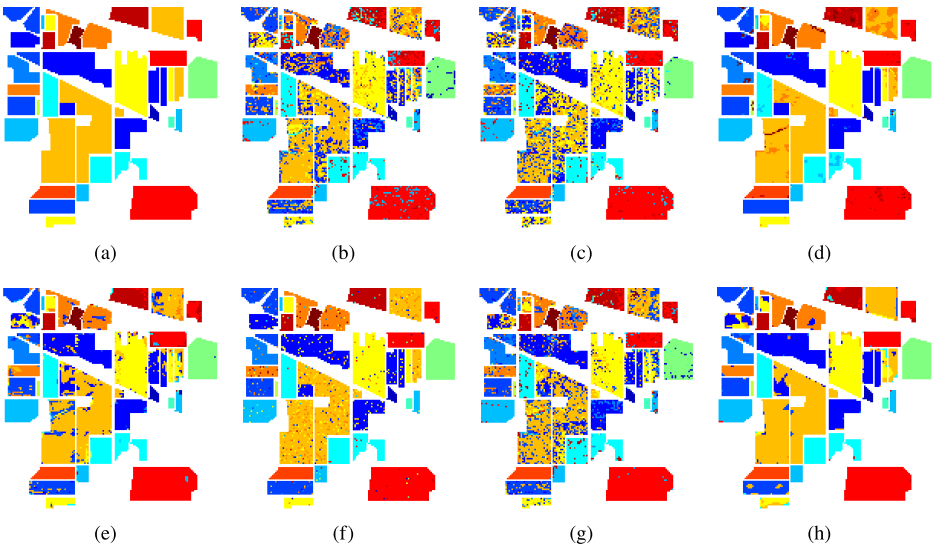


Fig. 7. Classification maps of the Indian Pines dataset by selecting 2% labeled samples for each class. Classification map obtained by (a) ground truth, (b) PCA + LP, (c) LRR + softmax, (d) KPCA + CE, (e) 2DCNN, (f) 3DCNN, (g) GCN, and (h) DS^{4}L.

2) In Tables V–VII, the AA of DS^{4}L reaches 97.88, 95.56, and 95.73, respectively. The evaluation results verify that our method can effectively improve the HSI classification accuracy by comparing related methods. It also proves that the regular constraint and irregular constraint in our method effectively improve the ability to capture representative subspace.
3) The accurate classification result is obtained under a small training sample, which shows that the model can alleviate the problem of insufficient labeled samples.

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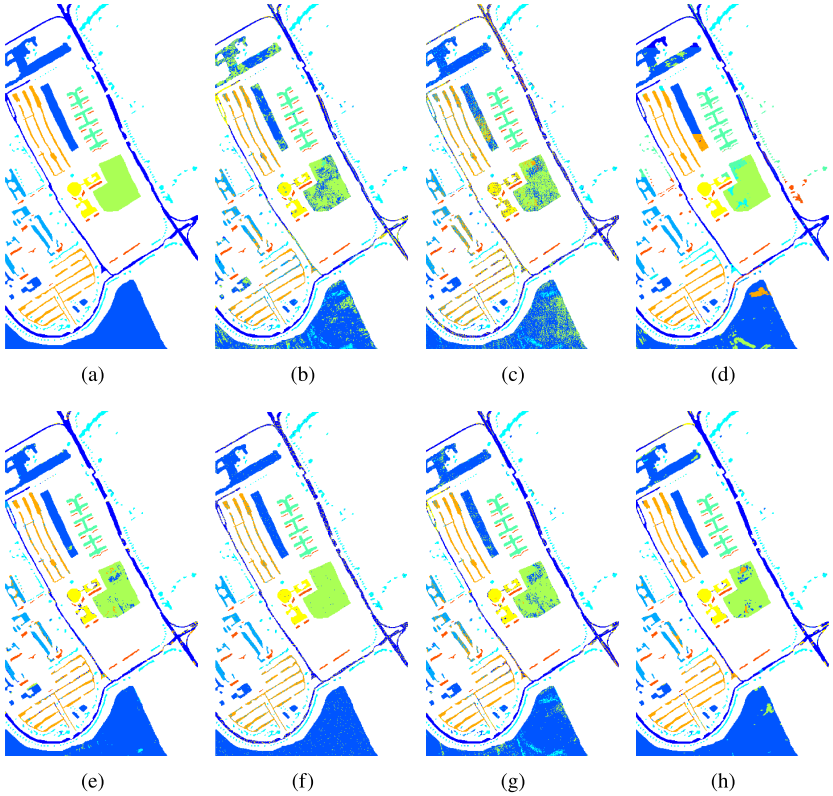


Fig. 8. Classification maps of the PaviaU dataset by selecting 2% labeled samples for each class. Classification map obtained by (a) ground truth, (b) PCA + LP, (c) LRR + softmax, (d) KPCA + CE, (e) 2DCNN, (f) 3DCNN, (g) GCN, and (h) DS^{4}L.

TABLE VII
OA OF DS^{4}L AND OTHER SUBSPACE-BASED METHODS

Dataset	SL	SL+RC	SL +IC	DS^{4}L
Salinas	90.88	93.28	92.15	**95.56**
Indian pines	89.10	91.21	91.78	**93.35**
PaviaU	93.60	94.32	94.63	**95.17**

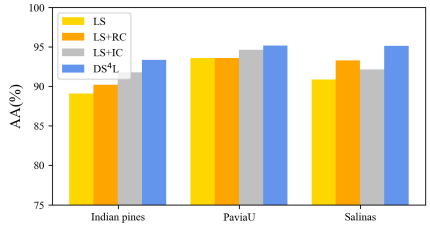


Fig. 9. Analyzing the importance of regular constraint and irregular constraint.

E. Analyzing the Effectiveness of Regular Constraint and Irregular Constraint

In this experiment, the other three variants of SL methods are compared with our DS^{4}L to prove the constraints' effectiveness in our model. These variants include the basic SL with multiscale input, the SL + *RC* that embeds the regular constraint in the basic SL, and the SL + IC that integrates irregular constraint in the basic SL. As shown in Table VII and Fig. 9, the SL + *RC* and the SL + IC make improvements in overall classification compared with the SL, which indicates that the regular constraint and the irregular constraint based on superpixel both enable the model to extract more effective subspaces.

F. Parameter Analysis

In DS^{4}L, the main adjustable parameters include hyperparameters, structural parameters, and other parameters. Hyperparameters mainly include learning rate, batch size, and regularization terms that are set to 10^{-5}, 128, and 10^{-3}, respectively, and then made adjustments according to each datasets training situation. For example, in the Salinas dataset, set the initial learning rate to 10^{-4} and then reduce it by 10^{-4} every 200 iterations for faster training and better performance. In the structural parameters, the out channels of BFE are set to 256, 768, and 1280 to increase the network's complexity.

For other parameters σ and λ_i and the main parameters of the superpixel clustering method *bandwidth*, the alternative parameter σ in the irregular constraint can be adjusted from $\{1, 10, 100, 200\}$. The *bandwidth* is used to control the centroid circle radius of the mean-shift model selected from the set $\{10^{-3}, 10^{-2}, \ldots, 10, 10^2\}$, and the final choice is based on the performance of the clustering results. The experimental results on three HSI datasets are shown in Fig. 10.

In formula (9), these three parameters λ_1, λ_2, and λ_3 are used to balance the weights of the three loss values in the objective function. λ_3 is fixed to 0.5. As shown in Fig. 11(a)–(c), the Salinas datasets parameters λ_1 and λ_2 are optimized to close to 0.5 and 0.1, showing that the regular

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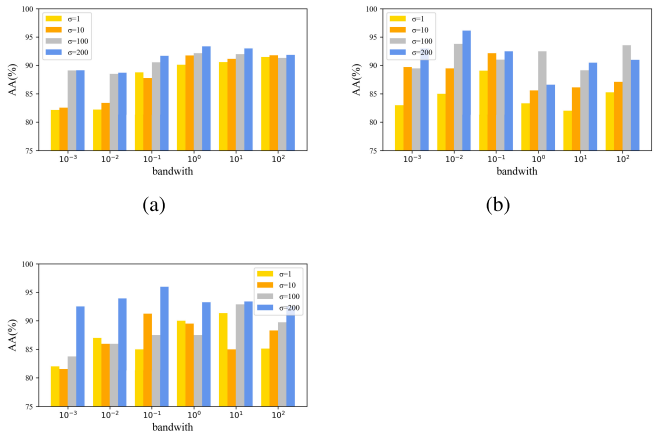


Fig. 10. Effect of irregular constraint's parameter σ and *bandwidth* obtained by (a) Salinas dataset, (b) Indian Pines dataset, and (c) PaviaU dataset.

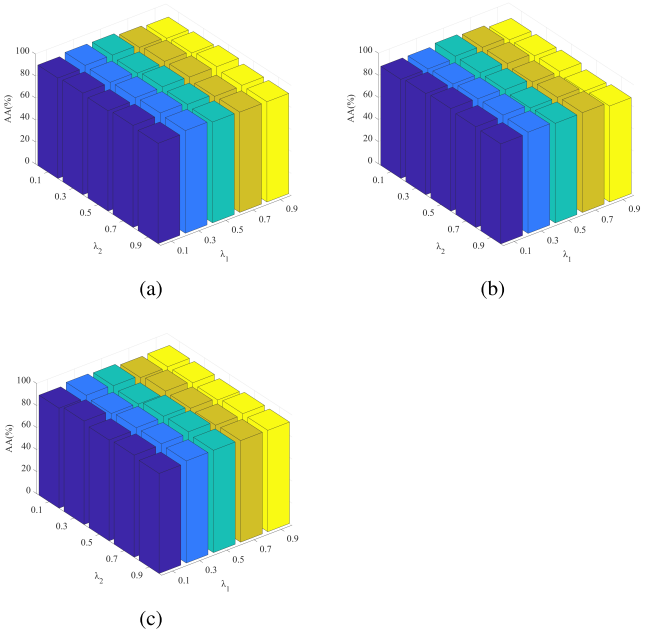


Fig. 11. Effect of the loss function's weight parameters. (a)–(c) Influences of λ_1 and λ_2 on the classification results for the Salinas, Indian Pines, and PaviaU datasets, respectively.

constraint plays a more obvious role here. For the Indian Pines and PaviaU dataset, the optimized parameters are closed to 0.3 and 0.5, 0.5, and 0.5, respectively. Therefore, the regular constraint and irregular constraint have the same effect on these two datasets.

IV. Conclusion

A new semisupervised HSI classification method called DS^{4}L is developed in this article. This model integrates the BFE network and the DSSL network to reduce the dependence on training samples and improve classification performance. The multiscale input is introduced in the BFE network to share multiscale subspace in the DSSL, which ensures the more sufficient and comprehensive information of subspace. In the DSSL network, we introduce the regular constraint and the irregular constraint to improve the representation of obtained subspace. In the regular constraint, multiscale patches are reconstructed to fully utilizing the spectral–spatial information of HSI. The irregular constraint with superpixel clustering is conducive to the accuracy of SL. The above contributions effectively improve the model's ability to extract effective subspace so that the DS^{4}L model only needs fewer samples for training. Besides, superpixel clustering optimized in each iteration introduces the contextual information between labeled and unlabeled samples to optimize the model in a semisupervised way. DS^{4}L obtains good results by training on a small number of labeled samples, indicating that our method can effectively alleviate the dependence on the number of samples and improve classification accuracy. For Salinas, Indian Pines, and PaviaU datasets, the OA reaches 97.88%, 95.56%, and 95.73%, respectively. The HSI classification results of these three datasets show the effectiveness of our method. Future work will continually combine the DS^{4}L model with more remote sensing data to develop products and conclusions of great importance in practical applications.

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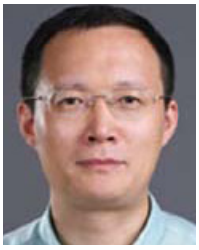
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